

1 Chapter 3.2

Problem 3. Prove that for $n \geq 2$

$$\prod_{j=2}^n \left(1 - \frac{1}{j^2}\right) = \frac{n+1}{2n}$$

Answer:

Proof. Base case: Using when $n = 2$ provides:

$$\underbrace{\left(1 - \frac{1}{2^2}\right) = \frac{4-1}{4} = \frac{3}{4}}_{LHS} = \underbrace{\frac{2+1}{2 \cdot 2} = \frac{3}{4}}_{RHS}$$

The inductive hypothesis is: $\prod_{j=2}^k \left(1 - \frac{1}{j^2}\right) = \frac{k+1}{2k}$. Starting with the left hand side.

$$\begin{aligned} \prod_{j=2}^{k+1} \left(1 - \frac{1}{j^2}\right) &= \left(1 - \frac{1}{(k+1)^2}\right) \prod_{j=2}^k \left(1 - \frac{1}{j^2}\right) \quad \text{break up terms} \\ &= \left(1 - \frac{1}{(k+1)^2}\right) \left(\frac{k+1}{2k}\right) \quad \text{use IHYP} \\ &= \left(\frac{(k+1)^2}{(k+1)^2} - \frac{1}{(k+1)^2}\right) \left(\frac{k+1}{2k}\right) \quad \text{algebra!!!} \\ &= \left(\frac{(k+1)^2 - 1}{(k+1)^2}\right) \left(\frac{k+1}{2k}\right) \\ &= \frac{(k+1)^2 - 1}{2k(k+1)} \\ &= \frac{k^2 + 2k + 1 - 1}{2k(k+1)} \\ &= \frac{k(k+2)}{2k(k+1)} \\ &= \frac{(k+1) + 1}{2(k+1)} \quad \text{Done} \end{aligned}$$

Therefore $S(k+1)$ is true, so $S(n)$ is true for all $n \geq 1$. □

Problem 7. Give a combinatorial proof: for $n \geq 1$

$$\sum_{j=1}^n j! < (n+1)!$$

Do so by asking a question and then performing a deliberate under- or over- count with one of the answers.

Problem 9. For the recurrence relation shown in (3.6), page 99, we proved $L_n < 2^n$ for $n \geq 1$.

1. Prove the tighter inequality $L_n \leq 1.7^n$. At what value of n should you start the induction?
2. What is so special about the number 1.7? Adjust your work in the first part to create the tightest bound you can.

2 Chapter 3.3

Problem 2. In each case, find a concise OGF for answering the question and also identify what coefficient you need.

1. How many ways are there to distribute 14 forks to 10 people so that each person receives one or two forks?
2. You can buy soda either by the can, or in 6, 12, 24 or 30 packs. How many ways are there to buy exactly k cans of soda?
3. How many ways are there to put a total postage of 75 cents on an envelope, using 3, 5, 10 and 12 cent stamps?
4. At the movies you select 24 pieces of candy from among five different types. How many ways can you do this if you want at least two pieces of each type?
5. How many solutions to $z_1 + z_2 + z_3 = 15$ are there, where the z_i are integers satisfying $0 \leq z_i \leq 8$?
6. How many ways are there to make change for a dollar using only pennies, nickels, dimes and quarters?

Problem 3. Find the coefficient of ...

1. x^{60} in $\frac{1}{(1-x)^{23}}$
2. x^k in $\frac{1+x+x^4}{(1-x)^5}$
3. x^3 in $\frac{x}{(1-x)^8}$
4. x^{50} in $(x^9 + x^{10} + x^{11} + \dots)^3$
5. x^{k-1} in $\frac{1+x}{(1-2x)^5}$

Problem 4. A professor grades an exam that has 20 questions worth five points each. The professor awards zero, two, four or five points on each problem. Find a concise OGF that can be used to determine the number of ways to obtain an exam score of k points.

Problem 7. Find the number of solutions to $z_1 + z_2 + z_3 + z_4 = 10$ where the z_i are nonnegative integers such that $z_1 \leq 4$, z_2 is odd, z_3 is prime and $z_4 \in \{1, 2, 3, 6, 8\}$.